

Customer Churn Analysis & Retention Insights

A decision-focused analysis of 64,374 customers, turning raw subscription data into clear churn signals, risk segments, and practical retention actions.

CUSTOMERS ANALYZED

64,374

Complete customer base

ATTRIBUTES

12

Customer & subscription fields

CHURN RATE

47.4%

Overall customers lost

REVENUE AT RISK

\$15.84M

Churned customer spend

ANALYSIS OBJECTIVE

Identify the behavioral, payment, and contract signals most strongly associated with customer churn — then translate them into actionable retention priorities.

ANALYSIS FOCUS

Customer profile

Usage & support

Payment behavior

Contracts & plans

Risk segmentation

Dataset

Customer Churn Dataset — Kaggle · muhammadshahidazeem

kaggle.com/datasets/muhammadshahidazeem/customer-churn-dataset

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Executive Churn Intelligence Dashboard

A consolidated view of customer profile, behavioral risk, payment friction and contract patterns

64,374

CUSTOMERS

30,493

CHURNED

47.4%

CHURN RATE

\$15.84M

REVENUE AT RISK

12,830

HIGH-RISK CUSTOMERS



KEY SIGNALS

Support calls and payment delays are the strongest warning signs · 7+ calls + 20+ late days → 88% churn · Monthly contracts churn 5 points above annual

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About the Dataset & Data Cleaning

Dataset

This report uses the **Customer Churn Dataset** from Kaggle, by **muhammadshahidazeem**. It has **64,374 customers** and This report examines the behavioral, payment and contract signals most closely associated with customer churn.

Link: <https://www.kaggle.com/datasets/muhammadshahidazeem/customer-churn-dataset>

COLUMN	DESCRIPTION
CustomerID	Unique ID for each customer
Age	Customer's age (18-65)
Gender	Male / Female
Tenure	Months the customer has stayed
Usage Frequency	How often they use the service each month
Support Calls	Number of support calls made
Payment Delay	Average days payments are late
Subscription Type	Basic, Standard, or Premium
Contract Length	Monthly, Quarterly, or Annual
Total Spend	Total amount spent (\$)
Last Interaction	Days since last activity
Churn	1 = churned, 0 = stayed

Data Cleaning

Before analysis, the data was checked in Python (Pandas):

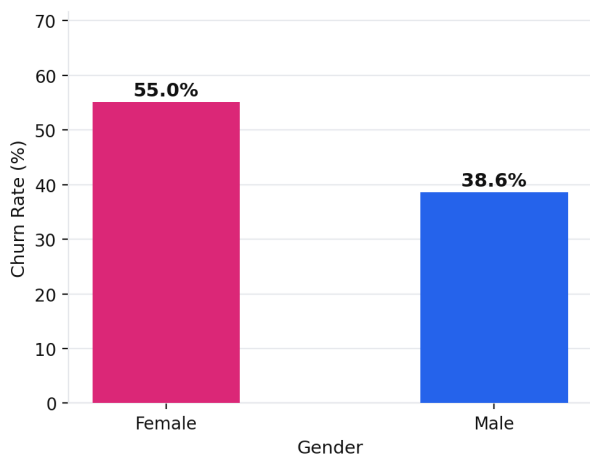
- **Missing values:** None found in any of the 12 columns.
- **Duplicates:** No duplicate rows found.
- **Data types:** Numbers and categories were checked and confirmed correct.
- **Outliers:** Checked with the IQR method — no unusual values found.
- **New columns:** Grouped Tenure, Support Calls, Payment Delay, and Last Interaction into ranges to make the charts easier to read.

Result: The data was already clean. No fixes were needed — just checks and a few new grouped columns for the charts.

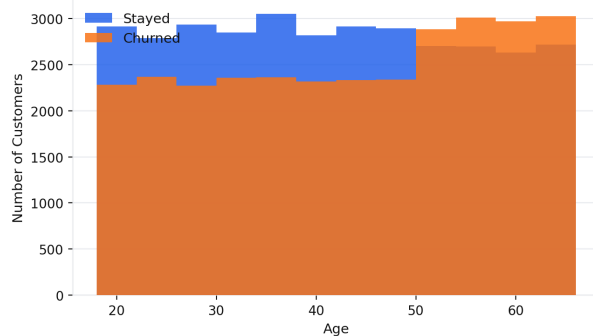
Who Is Churning?

This page looks at churn by gender, age, and spending.

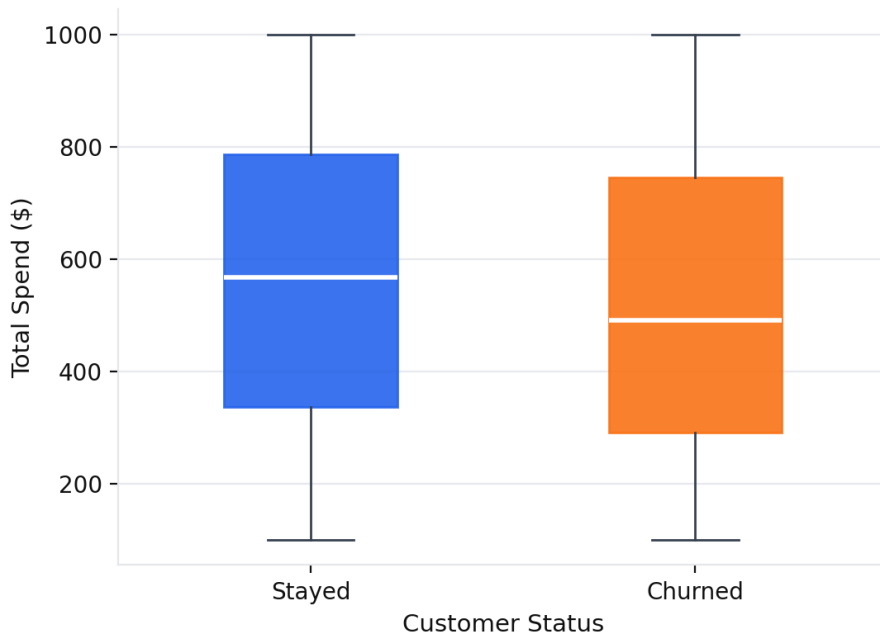
Churn Rate by Gender



Age Distribution by Churn Status



Total Spend: Stayed vs Churned

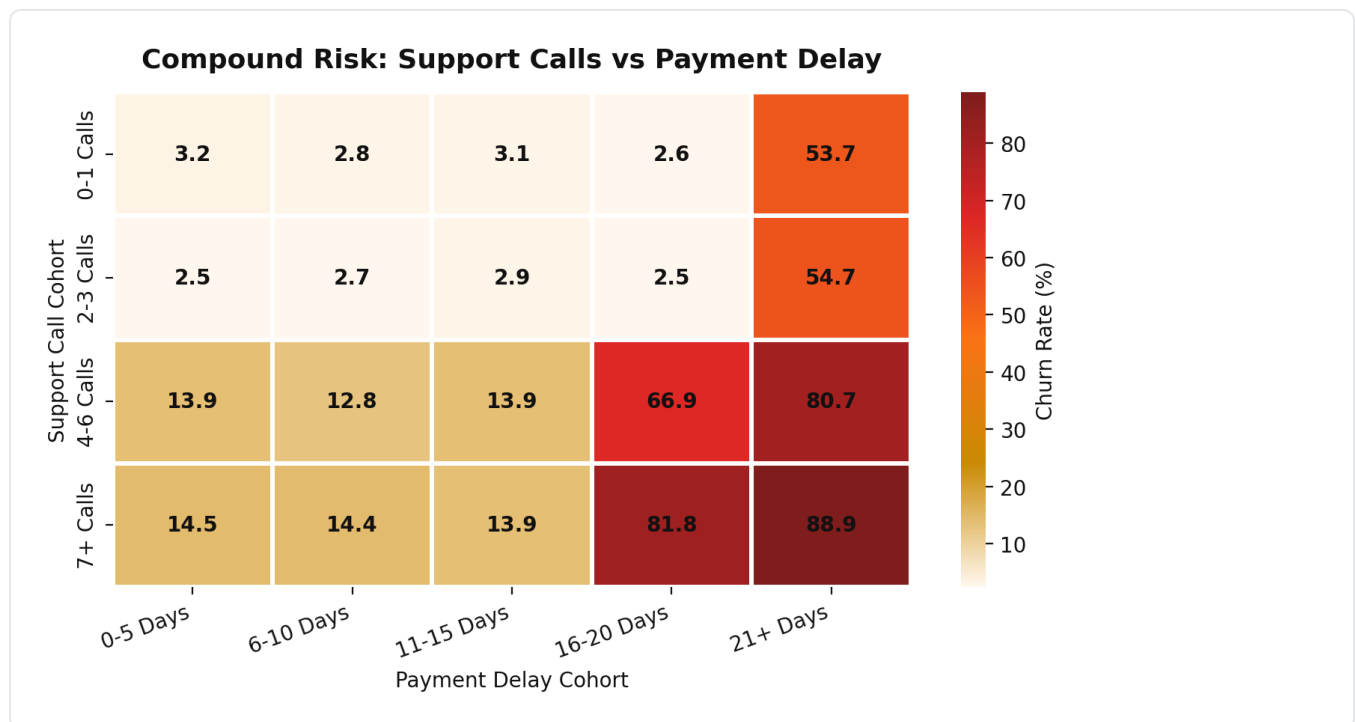
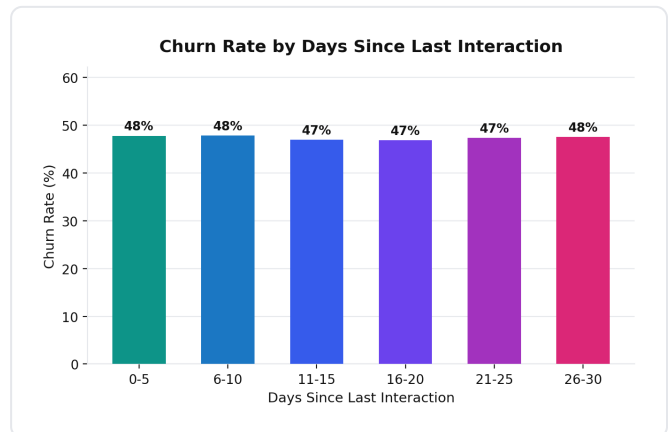
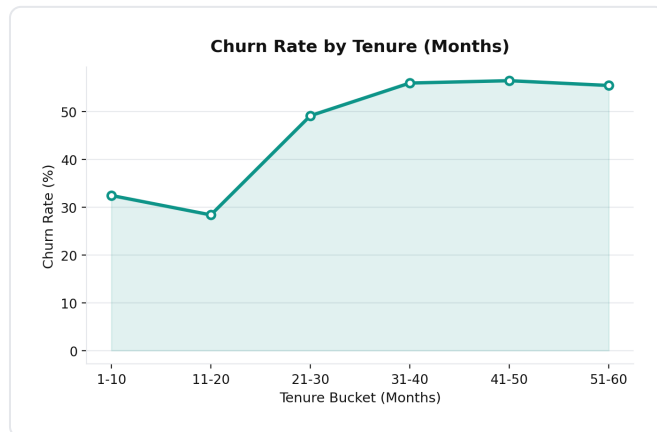
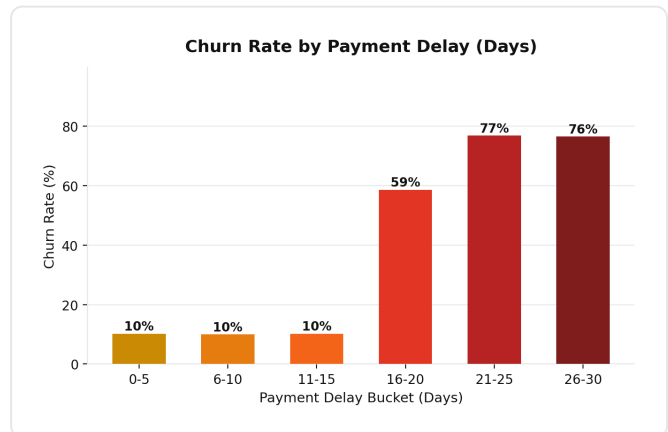
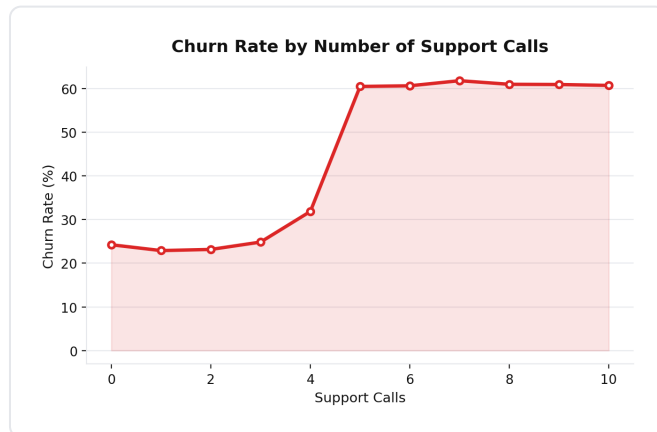


Gender matters most: Female customers churn at 55.0%, males at 38.6%. That's a big gap and worth looking into further.

Spend doesn't predict loyalty: Churned customers spent \$519 on average, retained customers spent \$561. High spenders leave too, so churn isn't just about low-value customers.

What Drives Churn?

These charts show how churn changes with support calls, payment delays, tenure, and recent activity.

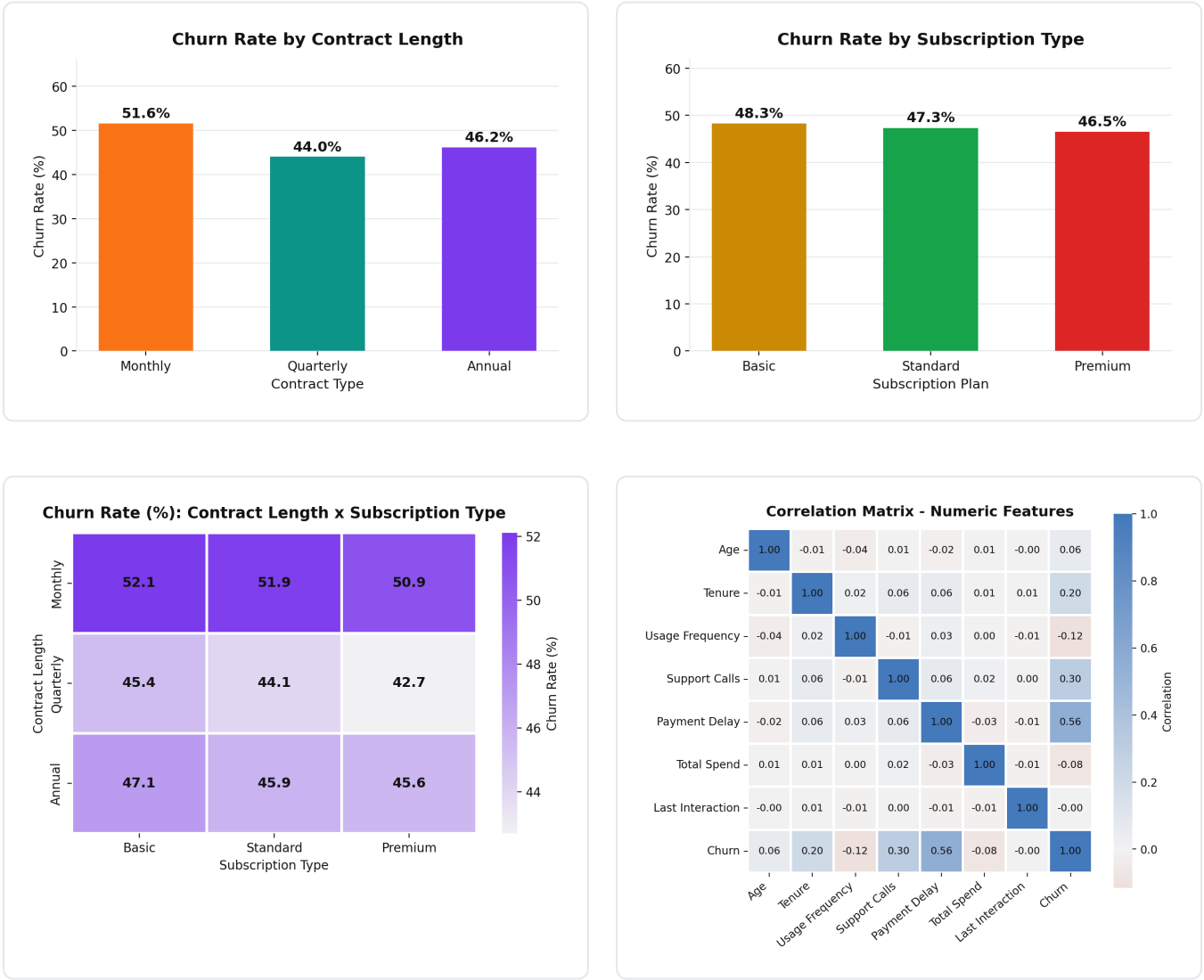


Support calls and late payments are the biggest warning signs. Churned customers made 6.4 support calls on average vs 4.5 for stayed customers, and paid 22 days late vs 12 days. The heatmap above shows the risk compounds fast: customers with 4+ calls AND 16+ late days churn far more than either factor alone. Customers with 7+ calls and 20+ late days (12,830 people) churn at 88%.

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Plans, Contracts & Correlations

This page compares churn across plans and contract types, and shows how the numbers relate to each other.



Monthly plans churn the most: 51.6% vs 44.0% (Quarterly) and 46.2% (Annual). Subscription tier (Basic/Standard/Premium) barely changes churn, so plan level alone doesn't predict who leaves.

The KPIs on the dashboard, written as DAX measures the way they'd be built in Power BI:

```
Churn Rate =  
DIVIDE( CALCULATE(COUNTROWS('Customers'), 'Customers'[Churn] = 1),  
        COUNTROWS('Customers') )  
  
Total Revenue = SUM('Customers'[Total Spend])  
  
Revenue at Risk = CALCULATE(SUM('Customers'[Total Spend]), 'Customers'[Churn] = 1)  
  
% Revenue at Risk = DIVIDE([Revenue at Risk], [Total Revenue])  
  
Avg Support Calls (Churned) =  
CALCULATE(AVERAGE('Customers'[Support Calls]), 'Customers'[Churn] = 1)  
  
High-Risk Customers =  
CALCULATE(COUNTROWS('Customers'),  
          'Customers'[Support Calls] >= 7,  
          'Customers'[Payment Delay] >= 20)
```

45%

REVENUE AT RISK

12,830

HIGH-RISK CUSTOMERS

88%

CHURN IN THAT GROUP

5pt

MONTHLY VS ANNUAL GAP

Recommendations

- **Focus on the high-risk group first:** customers with 7+ support calls and 20+ late days churn at 88%. Reach out to them first.
- **Push longer contracts:** Monthly customers churn 5 points more than Annual. Offer a discount to switch.
- **Look into the gender gap:** a 16-point gap between female and male churn is worth a closer look.
- **Fix late payments early:** automatic reminders or flexible billing dates could stop churn before it starts.
- **Don't ignore high spenders:** they churn too, so retention shouldn't only target low-value customers.

Turning Churn Data Into Action

Across 64,374 customers, churn is less about age or subscription tier and more about customer friction. Support volume, payment delays and shorter contracts consistently stand out as the clearest signals of retention risk.

01

Prioritize high-risk customers

Customers with 7+ support calls and 20+ late-payment days show 88% churn. Create an early-warning retention queue.

02

Reduce payment friction

Late payments rise sharply with churn. Use reminders, flexible billing dates and proactive outreach before delays become chronic.

03

Encourage longer contracts

Monthly customers churn at 51.6%, compared with 44.0% for quarterly and 46.2% for annual contracts.

EXECUTIVE TAKEAWAY

The strongest retention opportunity is not simply to target low-value customers. Build a risk-based program around support intensity and payment behavior, while using contract incentives to move customers toward longer commitments.